Applications of Linear Programming Regarding Intensity-Modulated Radiation Therapy With an Emphasis on Beam Intensity Optimization

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1 Introduction

The physiology of cancer has been extensively studied in response to the overwhelming number of cases and high mortality rate of cancer. In 2025, cancer accounted for approximately 618,000 deaths in the United States, with approximately 38.9% of people being diagnosed with cancer during their lifetime [1]. Treatment research has led to the development of intensity-modulated radiation therapy (IMRT), which is one of the most advanced techniques used to treat cancer and has been increasingly effective in recent decades due to advances in computers, imaging technologies, and planning improvements [2]. IMRT is used to deliver highly precise beams of radiation to target masses (tumors) while minimizing the dose experienced by surrounding, healthy tissue [3]. This is the fundamental goal of radiation therapy, as high doses of radiation have a strong relationship with tumor control; however, the damage caused to healthy tissue, especially immune cells, constrains the radiation dose [4][5]. For IMRT to be effective, a series of steps are used for its implementation. First, imaging is used to define the tumor and healthy tissue; CT scans are used to construct a three-dimensional model of the tumor by stacking multiple two-dimensional images [3]. Following this, there are three problems that fundamentally make up the process of IMRT: The angles of the radiation beams being projected into the tissue are determined, known as beam angle optimization (BAO); then, the optimal intensity of the radiation beams must be determined [6]. This is a key component to IMRT, as this allows a highly controlled radiation dose distribution. Finally, a machine called a multileaf collimator (MLC) is used to further conform the radiation dose by blocking beamlets using metal leaves that can vary in position to create the final beam shape [2]. Each of these three problems can be expressed as optimization problems with a linear objective function and constraints, so linear programming is an effective mathematical tool to apply. With regards to these three problems, the application of linear programming to beam intensity optimization will be examined thoroughly, and BAO and MLC optimization will be investigated briefly.

2 Beam Angle Optimization

The goal of BAO is to optimize the selection of beam angles such that the radiation dose delivered to healthy tissue is minimized while providing enough radiation to the tumor. To accomplish this, the model is constructed to select beam angles that minimize the number of beams needed while still delivering at least the minimum radiation dose to the tumor [6].

This is determined by selecting beams to be part of the solution by using binary decision variables. To model constraints on this objective function, a three-dimensional model of the tumor and surrounding tissue is formed where voxels are defined as either target, normal healthy tissue, or organs at risk (OARs). These constraints model the dose of radiation delivered to these voxels and set a minimum prescribed radiation dose to the target tissue and maximum doses for the normal tissues and OARs [6]. Modeling the BAO in this way results in a mixed-integer linear programming problem where the result selects a small set of beams that satisfy all dose constraints. Once beam angles are selected, beam intensity modulation can be investigated.

3 Multileaf Collimator Optimization

Once beam angles and intensities are optimized, the final problem is to optimize the position of leaves in the MLC. Beam intensity modulation optimization produces a fluence map, which is an idealized map of desired radiation dose; however, the linear accelerator that produces the radiation beam cannot conform the beam intensities with enough precision to execute the desired fluence map. To achieve the desired radiation distribution, an MLC must be used to shape the beam into delivering the radiation doses appropriately [2]. Therefore, the goal of MLC optimization is to minimize both the difference between idealized dose distribution (from the fluence map) and realized dose distribution (from the MLC settings) and the delivery time. The linear programming model uses leaf position and exposure time as decision variables, while the mechanical limitations of the MLC serve as the constraints. Specifically, leaf motor speed, minimum gap constraints, and other physical parameters of the MLC limit the MLC's ability to conform the beam shape. Ultimately, the precise delivery of radiation in coordination with the fluence map is limited by not only these constraints but also by the achievable precision of the physical aperture the MLC can create, as the amount and width of leaves affect the spatial resolution of the aperture. Solving this problem concludes the major linear programming calculations of IMRT treatment and transforms the treatment plan into a set of executable machine instructions.

4 Beam Intensity Modulation Optimization

4.1 Model

We consider the problem of finding an optimal way to administer radiation therapy to a cancer patient by modulating the intensity of beams selected by BAO. Suppose we have a matrix $D_{p,i}$ that describes the dose per intensity contributed by beam i to voxel p of some known organ scan. Then, if we define w_i to be the intensity of beam i, the dose (d_p) at a given voxel is

$$d_p = \sum_i D_{p,i} w_i.$$

At each voxel, there are bounds (b) on the acceptable dose, which are determined by medically-informed constraints about the treatment plan:

$$b_{\min,p} \leq d_p \leq b_{\max,p}. \tag{1}$$

There are also physical constraints on the attainable intensities of the beams. These constraints have an additional caveat that they need only apply when beam i is on. Thus, this constraint reads

$$w_i \leq M_i x_i, \tag{2}$$

where M_i is the maximum intensity of beam i, and x_i is a binary variable that denotes whether beam i is on.

One version of this model is to minimize the total dose of radiation

$$\min Z = \sum_i w_i \sum_p D_{p,i} \tag{3}$$

over the constraints defined by Equations 1 and 2.

4.1.1 Solving this model

We consider the two-dimensional version of this model, which can be understood as looking at a single plane of a three-dimensional organ scan. Suppose that we have I beams, each with a (centered) intensity profile represented by a Gaussian.

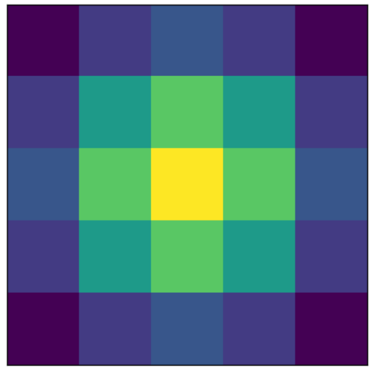


Figure 1: A sample intensity profile for a beam represented by 25 square pixels.

We then assume that the I unique beams are evenly distributed in a square across the target volume (visualized in Figure 2).

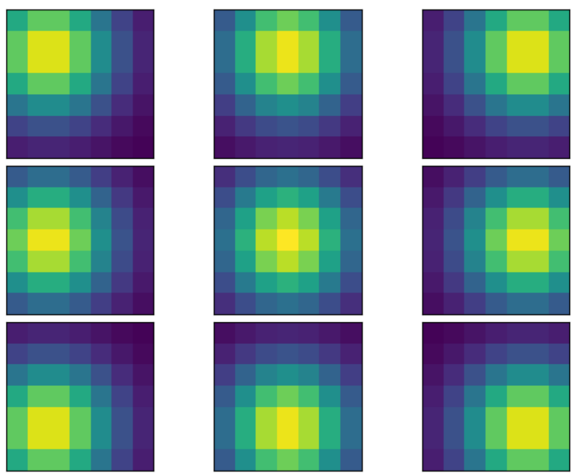


Figure 2: Intensity maps for a 7×7 target grid with 9 total beams.

For each beam, we flatten the two-dimensional data into a one-dimensional array, which corresponds to coefficients of the dose constraints in the model.

We now consider the bounds for the dose at a given voxel, p, in our two-dimensional scan of an organ. Given a scan of an organ, we assume that there is some known distribution of risk for the cells. Applying an arbitrary thresholding model, we generate a map of cancerous cells, as shown in Figure 3, and determine the maximum and minimum dose for each category.

Choosing the maximum intensity for each of the beams to be 1, we have the following

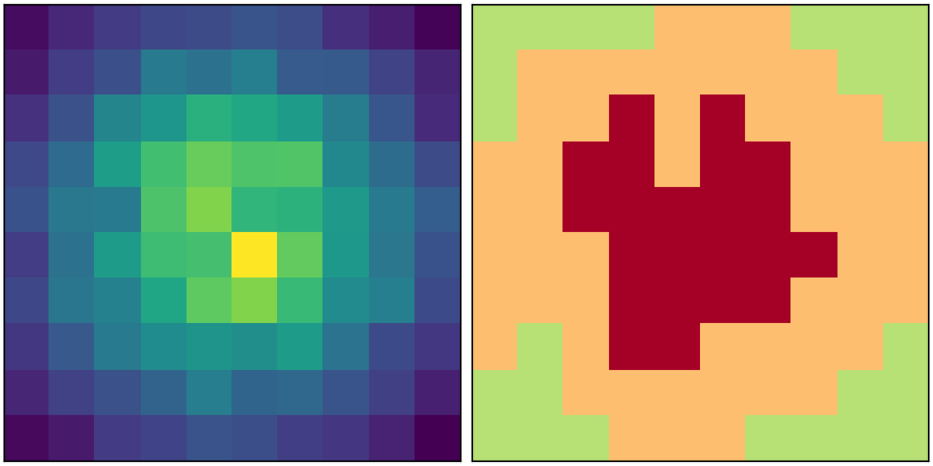


Figure 3: An example scan of a tumor (left) with regions of high, medium, and low risk corresponding to red, yellow, and green, respectively (right).

linear programming problem:

$$\min Z = \sum_i w_i \sum_p D_{p,i}$$

$$b_{min,p} \le \sum_i D_{p,i} w_i \le b_{max,p}$$

$$w_i \le x_i$$

where the w_i are greater than zero.

4.1.2 Numerical simulations

Applying the exact approach outlined above in our python code, the problem is infeasible for every combination of parameters that we tested. This suggests the need for realistic data for each of the constraints, but can also be understood by looking closer at the beam intensity profiles.

Considering a tumor region similar to the one in Figure 3, we use 9 beams and generate the beam profile in Figure 4. Because our current model scales the beam intensity profile with one coefficient, there is no way to achieve the dose bounds for all voxels, and thus the problem is infeasible.

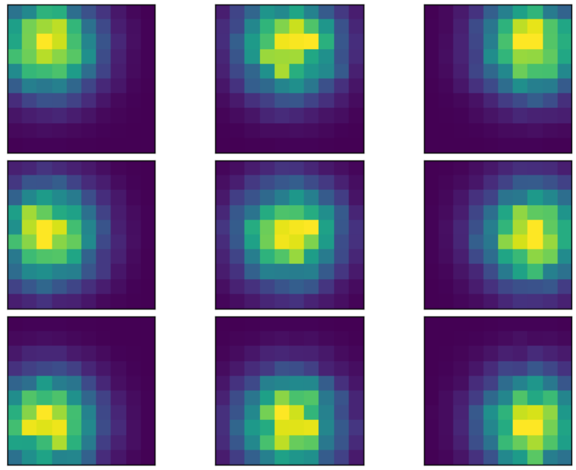


Figure 4: Example 9 beam array for a 10×10 voxel tumor scan.

The initial reason to consider this approach was to look at a situation that is realistic for an array of imperfect beams incident on the target volume. The analysis thus far suggests that the current method is not appropriate and we must revise our model slightly.

By decreasing the spread of each beam and increasing the total number of beams considered, we can examine a feasible linear programming problem. For an $N \times N$ organ scan, we need N^2 beams with nearly pixel-level precision to deliver the necessary dose and stay within the constraints.

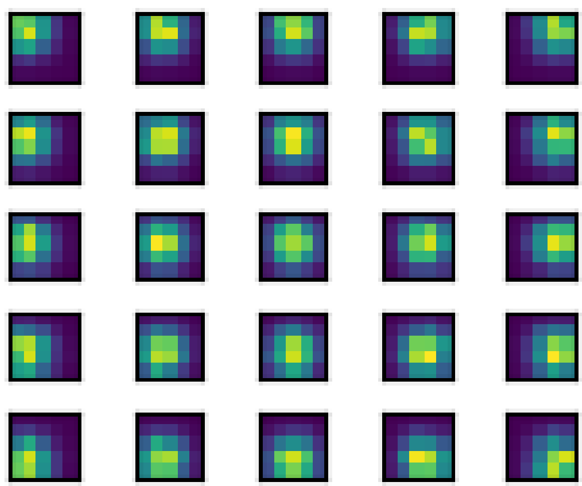
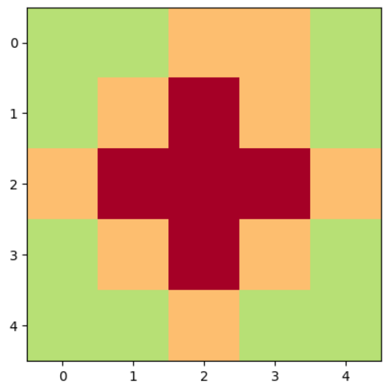
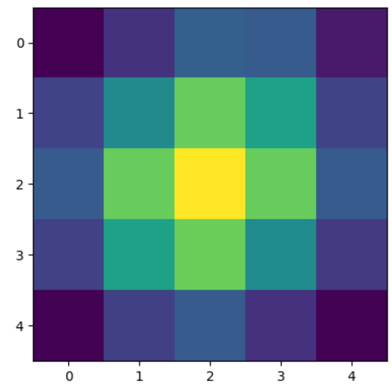


Figure 5: A target volume with 25 total pixels requires 25 beams to deliver the necessary doses.

With all of these constraints in the model, we find the appropriate intensity for each beam and construct a graphic of the proposed dose. As we see in Figures 6a and 6b, the tumor scan and dosage scan clearly agree on regions of highest, medium, and lowest risk/dose.



(a) Thresholded organ scan for solved problem.



(b) Total proposed radiation intensity map.

5 Conclusion

5.1 Results

Our numerical simulation has demonstrated that the success of IMRT beam intensity modulation optimization is highly dependent on the resolution of beam intensity maps. This emphasizes a critical mathematical limitation to the application of linear programming in IMRT: there must be enough degrees of freedom to create a feasible solution satisfying the dose constraints' lower and upper bounds. However, our model was able to construct a feasible solution for an $N \times N$ organ scan using N^2 beams. This result shows linear programming can be used to find solutions for IMRT beam intensity modulation optimization, supporting our hypothesis that linear programming is an effective tool for this problem. Nonetheless, using N^2 beams to solve the $N \times N$ organ scan is an unreasonably large number of beams to use for real clinical application, and so the model needs significant refinement to be used in real IMRT treatment. Furthermore, this model can be improved by developing more constraints based on physical and medical realities (a whole section of [7] examines such constraints).

5.2 Future directions

Our model functions as a proof-of-concept for the application of linear programming in IMRT, yet the many simplifying assumptions of our model make it unfit for clinical application of IMRT. One of the most significant assumptions this model makes is the idealized immobility of the target tissue; however, small changes in patient position due to breathing, other organ movements, or changes to patient anatomy create uncertainty in the dose distribution to tissue [8]. To account for this uncertainty, robust optimization is used in real-world applications of these models to ensure the solution remains feasible under these small changes. Mathematically, these movements cause fluctuation in the dose-per-intensity matrix, $D_{p,i}$. Robust optimization would allow a solution to remain feasible and effective after these changes to $D_{p,i}$ occur.

Another major simplifying assumption was to exclude any consideration for time. Typically, radiation dose delivery time is tightly constrained and minimized using linear programming to reduce the variability in the system. Furthermore, dose administration time must be modeled for subsequent MLC optimization problem. This model was not sophisticated enough to consider this constraint; however, a more advanced model would include this.

Finally, our model worked with a uniform intensity of each beam; however, more advanced approaches divide each beam into several beamlets that are individually tunable. This technique results in a more comprehensive and precise treatment that requires fewer beams. This sort of improvement could address the feasibility issue that we encountered when modeling the beams with a wider spatial extent. With more time, our model can be refined to represent each beamlet, which would increase the degrees of freedom in the model, making it easier to satisfy the radiation dose bounds while keeping the total beams to a minimum.

5.3 Reflection

While we worked with artificially generated coefficients and a smaller model scale than seen in real clinical work, this project serves as a proof-of-concept. This work describes the application of linear programming to the three significant problems of IMRT, BAO, beam intensity modulation optimization, and MLC optimization. More specifically, this project detailed the application of linear programming to optimizing beam intensities that allow clinical IMRT to effectively control cancer using high radiation doses to target tissue while maintaining safe levels of radiation delivered to healthy tissues. We anticipated that linear programming could be used to model IMRT; however, our model highlights a pressing issue: working with real data. Despite the major assumptions and limitations of our work, it shows there are many opportunities to use linear programming in this problem, and more detailed models prove to be useful tools for cancer treatment.

References

[1] National Cancer Institute. (2025, May 7). *Cancer statistics*. U.S. Department of Health and Human Services, National Institutes of Health. https://www.cancer.gov/about-cancer/understanding/statistics

[2] Cho, B. (2018). Intensity-modulated radiation therapy: a review with a physics perspective. *Radiation Oncology Journal, 36*(1), 1-10. https://pmc.ncbi.nlm.nih.gov/articles/PMC5903356/

[3] Taylor, A., & Powell, M. E. B. (2004). Intensity-modulated radiotherapy—what is it? *Cancer Imaging, 4*(2), 68-73. https://pmc.ncbi.nlm.nih.gov/articles/PMC1434586/

[4] ScienceDirect. (n.d.). *Cancer control*. https://www.sciencedirect.com/topics/nursing-and-health-professions/cancer-control

[5] Jin, J., Hu, C., Xiao, Y., Zhang, H., Paulus, R., Ellsworth, S. G., Schild, S. E., Bogart, J. A., Dobelbower, M. C., Kavadi, V. S., Narayan, S., Lyengar, P., Robinson, C., Greenberger, J. S., Koprowski, C., Machtay, M., Curran, W., Choy, H., Bradley, J. D., & Kong, F. S. (2021). Higher Radiation Dose to the Immune Cells Correlates with Worse Tumor Control and Overall Survival in Patients with Stage III NSCLC: A Secondary Analysis of RTOG0617. *Cancers, 13*(24), 6193. https://www.mdpi.com/2072-6694/13/24/6193

[6] Huang, P., Shang, J., Xie, X., Hu, Z., Liu, L., & Yan, H. (2024). Applying mixed-integer linear programming to the non-coplanar beam angle optimization of intensity-modulated radiotherapy for liver cancer. *Quantitative Imaging in Medicine and Surgery, 14*(8), 5789-5802. https://pmc.ncbi.nlm.nih.gov/articles/PMC11320540/

[7] Lee, E. K., Fox, T., & Crocker, I. (2003). Integer Programming Applied to Intensity Modulated Radiation Therapy Treatment Planning. *Annals of Operations Research, 119*, 165-181. https://www.opt.math.tugraz.at/ cela/Vorlesungen/MathOptSS13/Miniprojekt/radiation2Lee.pdf

[8] Sevilla, A. C., Cabal, G., Wahl, N., Puerta, M. E., Rivera, J. C. (2025). A robust optimization model for intensity-modulated radiotherapy: Cheap-Minimax. *Medical Physics, 52*(5), 3360-3376. https://pmc.ncbi.nlm.nih.gov/articles/PMC12082758/